

WERL, Germany

WMO: 104240

Lat: 51.5764N

Lon: 7.8881E

Elev: 86

StdP: 100.29

Time Zone: 1.00 (EUC)

Period: 03-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.3	-6.1	-12.3	1.3	-8.2	-9.3	1.7	-5.0	13.5	8.5	11.8	8.2	2.0	60	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	29.8	19.2	27.6	18.7	25.6	17.9	20.5	26.8	19.7	25.4	18.8	24.1	3.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.5	22.5	17.6	12.7	21.7	16.8	12.1	20.9	59.2	26.8	56.5	25.5	53.7	24.1	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.6	7.5	-12.3	34.0	4.7	2.1	-15.7	35.5	-18.4	36.7	-21.1	37.9	-24.5	39.4		
(5)			-12.7	21.9	4.7	0.7	-16.0	22.4	-18.8	22.8	-21.4	23.2	-24.8	23.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	2.5	2.6	5.6	10.1	13.0	16.0	18.5	18.1	15.1	10.9	7.1	2.9		
	DBStd	6.94	4.97	4.44	4.36	3.77	3.51	3.33	3.31	2.75	3.61	3.85	3.77	4.27		
	HDD10.0	1005	235	208	145	44	12	1	0	0	4	35	101	221		
	HDD18.3	3053	492	440	394	247	168	84	39	38	108	229	336	478		
	CDD10.0	1097	2	2	9	48	105	182	263	251	157	64	14	1		
	CDD18.3	104	0	0	0	0	3	15	44	30	11	1	0	0		
	CDH23.3	1006	0	0	0	12	39	146	446	264	96	2	0	0		
	CDH26.7	274	0	0	0	0	6	29	146	78	15	0	0	0		
(14) Wind	WSAvg	3.4	4.4	3.9	3.8	3.1	3.0	2.9	2.9	3.0	3.0	3.2	3.8	4.1		
(15) Precipitation	PrecAvg	835	75	63	61	47	62	71	87	83	71	69	75	74		
	PrecMax	1087	164	147	143	84	117	120	184	168	145	211	129	185		
	PrecMin	620	12	17	8	1	27	21	24	30	18	22	0	25		
	PrecStd	123	37	34	34	23	24	25	35	40	32	36	33	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	14.2	18.6	25.1	27.3	29.9	33.4	31.7	28.7	22.6	17.1	13.2		
		MCWB	10.7	10.7	11.8	15.3	19.1	20.3	19.8	19.8	18.2	16.1	12.7	10.6		
	2%	DB	11.9	11.6	16.0	22.2	24.1	27.1	30.7	28.8	26.3	20.2	15.3	11.1		
		MCWB	10.2	8.9	10.8	14.5	16.5	19.2	19.4	18.4	18.2	15.6	12.0	9.1		
	5%	DB	10.3	10.0	14.1	19.7	22.2	24.7	28.1	26.2	23.7	18.2	13.9	9.7		
		MCWB	8.6	8.0	10.3	13.3	15.1	18.1	18.9	18.0	17.5	14.5	11.3	8.0		
	10%	DB	8.8	8.7	12.1	17.3	20.2	22.9	25.8	23.8	21.2	16.7	12.4	8.6		
		MCWB	7.3	7.0	8.8	12.0	14.6	17.1	18.3	17.2	16.3	13.7	10.4	7.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.0	13.0	16.1	19.9	21.5	21.9	21.1	20.2	17.2	13.3	11.1		
		MCDB	12.6	13.3	16.4	22.4	25.7	27.7	27.7	29.2	24.9	21.0	15.8	12.7		
	2%	WB	10.2	9.5	11.8	14.8	18.1	20.1	20.6	19.6	19.0	16.0	12.4	9.2		
		MCDB	11.6	11.1	15.2	21.1	22.8	25.2	27.3	25.7	24.9	19.5	14.8	10.7		
	5%	WB	8.8	8.3	10.5	13.7	16.4	18.9	19.8	18.7	17.9	15.0	11.6	8.2		
		MCDB	10.3	9.8	13.4	18.7	20.0	23.4	26.1	24.1	22.4	17.7	13.8	9.6		
	10%	WB	7.3	6.9	9.1	12.5	15.0	17.9	18.9	17.8	16.7	14.0	10.5	7.0		
		MCDB	8.7	8.6	11.8	16.8	19.1	22.0	24.5	22.7	20.7	16.5	12.5	8.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	4.8	7.1	9.7	9.6	9.8	10.1	8.9	8.7	7.3	5.0	4.3		
		MCDBR	4.7	6.1	10.9	13.9	13.5	13.3	14.7	13.7	13.2	9.2	6.2	4.6		
	5% WB	MCWBR	4.1	4.3	6.6	7.6	7.2	6.7	5.4	5.4	6.4	5.3	4.3	4.1		
		MCDWR	4.7	5.1	9.4	12.5	11.3	11.4	12.6	10.7	11.3	8.1	5.5	4.6		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.344	0.377	0.412	0.416	0.416	0.427	0.421	0.390	0.361	0.340	0.318		
		taud	2.387	2.346	2.278	2.222	2.242	2.262	2.238	2.280	2.354	2.419	2.399	2.379		
	Ebn at Noon		689	771	811	823	838	840	823	807	790	742	654	625		
		Edh at Noon	61	83	107	127	130	129	130	119	99	77	61	53		
(44) All-Sky Solar Radiation	RadAvg		0.71	1.38	2.49	4.01	4.85	5.23	4.98	4.24	3.08	1.81	0.83	0.54		
	RadStd		0.10	0.25	0.36	0.47	0.47	0.46	0.65	0.39	0.39	0.17	0.12	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page